INCH-POUND

A-A-52486
January 18, 1994
Superseding
MIL-M-45907A(AT)
13 December 1984

COMMERCIAL ITEM DESCRIPTION

MOUNT, SHIPPING CONTAINER, RESILIENT: SHOCK AND VIBRATION DAMPING

The General Services Administration has authorized the use of this commercial item description (CID) as a replacement for MIL-M-45907A(AT), which is canceled.

ABSTRACT

This CID covers resilient mounts used to support major components (such as engines, transmissions and axles) when transported in reusable metal shipping containers.

SALIENT CHARACTERISTICS

a. Description. Mounts furnished under this CID are intended for use in securing major components of military vehicles (engines, transmissions, etc.) in reusable shipping containers. The mount functions as a shock absorber and vibration damper between the container and the component mounted therein. The specific shape, dimensions, and characteristics of each mount shall be defined by its drawing (see note a).

b. Materials. Materials shall be as specified herein and in the applicable resilient mount drawing. All material shall be uniform in quality and free from defects which could adversely affect performance or serviceability of the mounts. The use of recovered materials made in compliance with regulatory requirements is acceptable providing that all requirements of this CID are met (see note c).

1. Elastomers. The elastomeric material comprising the resilient elements of the mounts shall be an appropriate elastomer compounded to meet the specified performance requirements.

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000.

AMSC N/A FSC 5340

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2. Steel plates and studs. The mechanical properties of the steel used in the manufacture of mount plates and studs shall be such as to enable these components to withstand, without damage, the stresses applied in connection with the performance requirements specified herein.

3. Threaded parts. Mounting stud threads and tapped holes shall be in accordance with commercial practice. The thread form, number of threads per inch, and class of fit shall be as specified on applicable drawings.

4. Design and construction. The design and construction of each resilient mount shall be in accordance with its drawing and this CID, such that it properly performs its intended function. The resilient mounts shall be of the type designed for shear loading in the vertical plane.

5. Interchangeability. All mounts furnished under this specification shall be dimensionally and functionally interchangeable with all other such mounts having the same part number.

6. Dimensions. All resilient mount dimensions shall be in accordance with the applicable drawing.

7. Bond. All metal components of the mounts shall be completely bonded to the resilient element with which each makes contact.

c. Performance.

1. Static load and proof load deflection. The mounts shall exhibit static load and proof load deflection characteristics (spring rates) in accordance with the values specified in the applicable drawing. Suggested test mounting configurations are shown on figures 1 and 2. This shall be demonstrated by testing two initial production samples from the first batch of 1000 mounts and two from each subsequent batch of 2000. Prior to recording test data, the mounts shall be statically deflected twice in the shear direction a distance as recommended by the mount manufacturer for that mount. During a third deflection, load readings shall be recorded at one inch deflection. The loading rate for the static load shall be 0.5 to 1 inch per minute until the specified maximum deflection is reached. The loading rate for the proof load shall be 25 $\pm$ 5 inches per minute until the specified maximum deflection is reached and held at maximum deflection for 30 to 60 seconds. The recorded loads for both static and proof loads shall be within drawing ranges. During and after testing, the sample mounts shall show no evidence of performance deterioration and no damage, such as cuts, tears, bond separation, or metal deformation.

2. Dynamic performance. A pair of mounts from the first batch of 1000 mounts and a pair from each subsequent batch of 2000 mounts shall perform as specified following, while supporting a load as specified in the applicable drawing.

a. Vibration resistance. When tested on a set up similar to that shown in figure 2, the mounts shall be capable of withstanding, without damage, vibration at the frequencies, amplitudes and for the duration of periods specified in salient characteristic (SC) c2b and c2c. The dummy load shall be equal to the weight the mounts are required to support statically in the specific container application. Vibration shall be conducted only in the direction parallel to the vertical shear axis of the mounts. Vibration pickups shall be mounted on both the dummy load and the input platform.

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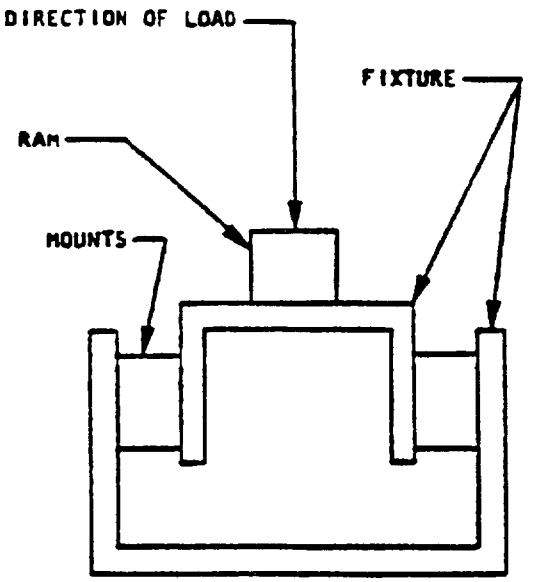


FIGURE 1. Static load deflection.

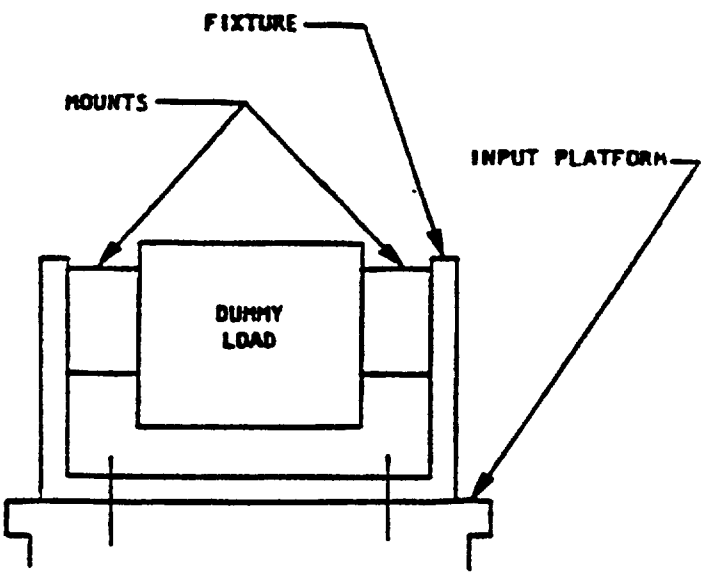


FIGURE 2. Vibration test.

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b. Resonance vibration frequency. When tested on a set up similar to that shown in figure 2, and after being subjected to vibration at 0.06 inch double amplitude over a frequency range of 2 to 60 cycles per second (cps), the mounts shall be such as to preclude a natural or resonant vibration frequency within the carrier (railway or motor freight) vibration frequency range of 2 to 7 cps.

c. Resonance vibration fatigue. When tested on a set up similar to that shown in figure 2, the mounts shall be capable of withstanding, without damage, a total duration period of 21 ± 2 minutes when exposed for 3 sweeps to the frequency range of 2 to 60 cps and return to 2 cps. Each sweep time shall be 7 ± 1 minutes. Vibration shall be steady state for 15 minutes when the resonant frequency is above 20 cps. Intermittent dwell testing is permitted to avoid overheating of elastomeric mounts.

3. Shock attenuation. With the mounts and dummy load forming a center of gravity system along the vertical axis and an accelerometer mounted on the dummy load directly over the center of gravity, the shock test shall be carried in a set up similar to the one in SC c2a. The entire assembly shall be dropped flat to a hard non-yielding surface three times from a height of 12 ± 1/2 inches. The mounts shall evidence no damage and no shock greater than 15 g's shall be registered.

4. Durability. The mounts shall be capable of withstanding, without evidence of damage, 50 drops from a height of 18 ± 0.5 inches. The contractor shall perform appropriate drop tests to verify conformance to this requirement.

d. Environmental requirements.

1. Accelerated aging. After exposure to accelerated temperatures of 158 to 160 degrees Fahrenheit (°F) for a period of 7 days, two mounts from the first batch of 1000 and two from each subsequent batch of 2000 shall be allowed to cool to room temperature and then shall meet the requirements of SC c1 within 20 percent of the original room temperature values. The contractor shall devise and perform accelerated aging at temperature tests to verify conformance to this requirement.

2. Low temperature. The mounts shall evidence no damage and no shock greater than 35 g's shall be registered when the entire assembly, after being subjected to a temperature of -38° to -40°F for 16 hours, is still or immediately after being removed from the cold chamber and subjected to SC c3 test.

3. High temperature. The mounts shall evidence no damage and no shock greater than 15 g's shall be registered when the entire assembly, after being subjected to a temperature of 158° to 160°F for not less than 4 hours, is removed from the ventilated oven and subjected to SC c3 test.

e. Identification and markings. The mounts shall be marked for identification in accordance with the drawing for the specific part number. The cure date (month and year) and the manufacturer's identification code (CAGE) shall also be molded into the resilient element. The mounts shall be visually examined during inspection to verify conformance to the identification requirements herein.

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QUALITY ASSURANCE PROVISIONS

a. Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

b. First article inspection. Unless otherwise specified (see note a), a first article sample shall be inspected to verify conformance to the salient characteristics in this CID. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply items that are fully representative of those inspected as a first article sample.

c. Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and that the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

PRESERVATION, PACKAGING, PACKING, LABELING, AND MARKING

Preservation, packaging, packing, labeling and marking for the desired level shall be as specified in the contract (see note a).

NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

a. Ordering data. Acquisition documents must specify the following:

1. Title, number, and date of this CID.
2. Title, number and revision letter of the applicable engineering drawing for the resilient mount.
3. Arrangements for first article inspection or whether first article is not required.
4. Applicable level and packaging requirements.

b. Cross-reference data. Resilient mounts conforming to this CID are interchangeable/substitutable with resilient mounts conforming to MIL-M-45907A(AT) dated 13 December 1984.

c. Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical.

MILITARY INTERESTS:

Custodian
Army - AT

Review activities
DLA - IS
DISC - EHC

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FSS

Preparing activity:
Army-AT

(Project 5340-2093)

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